An Optimized LTE-Based Technique for Drone Base Station Dynamic 3D Placement and Resource Allocation in Delay-Sensitive M2M Networks

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Abstract—Drones are expected to facilitate extending wireless networks' access for both human users and smart machine-type-communication devices (MTCDs) with strict and diverse quality of service (QoS) requirements. In this paper, we propose an optimal solution for the dynamic placement of an LTE drone-mounted base station to maximize the coverage of the MTCDs deployed over a large geographical area. The resulting technique jointly optimizes the drone's 3D positioning to maximize coverage and allocates the network resources in such a way that gives high priority to the delay-sensitive machine-to-machine (M2M) traffic. This optimization algorithm determines the optimal bound of the solution. Since it cannot be used in real-time operations due to its computational complexity, we also introduce a heuristic technique that offers a near-optimal solution with much reduced complexity. We conduct several simulation evaluations to assess the proposed techniques. These results are compared to those of other drone placement approaches. The comparisons show that the proposed techniques offer significantly better results in the communication coverage while fulfilling the diverse QoS requirements of the deployed M2M network. Moreover, the heuristic-based technique is shown to succeed in finding solutions close to the optimal bound with a considerable reduction of complexity over the exact algorithm.

Index Terms—Drone base station, UAV-assisted network, exact optimization, machine-to-machine communication, LTE coverage

1 Introduction

Recently, the demand for using unmanned aerial vehicles (UAVs), also known as drones, has increased dramatically in both civilian and military applications due to their high mobility and low cost [1]. The use of drones is also expected to result in significant advances in future wireless communication systems. For instance, UAVs can be used to extend wireless network coverage to remote areas with no communication services or to disaster areas that might have damaged network infrastructure [1]. The UAV-aided communication system is usually divided into three typical functions, namely, UAV-aided ubiquitous coverage, UAV-aided relaying and UAV-aided information dissemination and data collection [1]. In the UAV-aided ubiquitous coverage function, drones are used to enhance the coverage or the performance of an existing communication infrastructure that might encounter overloading or malfunctioning situations. In the relaying function, drones are exploited as gateways where they can connect network clusters to the core network. In the data collection/dissemination function, the drones act as data mules where they gather/spread information from/to remote network nodes that are not directly connected to the first instance of the network infrastructure. The different functions in which UAVs can be used to support communication networks are considered a major enabler of the massive Internet of things (IoT) use cases in 5G networks. This massive IoT deployment involves the LTE technology as a fundamental component of the 5G system to support the IoT communication requirements [2]. In particular, the machine-type-communication devices (MTCDs), which are basically deployed in harsh/remote environments, can benefit from such aerial communication functions along with the LTE technology's IoT-support releases. In machine-to-machine (M2M) applications, the direction of data transfer of the MTCDs' traffic is commonly in the uplink direction [3]. This results in contention in using the limited network resources, especially with the strict constraints on the data transmission deadlines in many of the M2M applications. With the aid of the controlled flying positions of UAVs, the machine-type communications (MTC) are expected to benefit from the aerial-supported communications due to the ability to adapt the UAV positions in accordance with the deployed MTCDs' connectivity on the available communication channels which supports the functions of the uplink schedulers.

Therefore, some previous research efforts considered the deployment of drones as communication access providers. Accordingly, drones are proposed to act as base stations or relay nodes. These studies are mainly focused on the placement problem of the airborne platforms along with some other

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communication factors such as the transmission power and the system throughput. Hence, the problems formulated in the previous research work aim at maximizing either the number of covered users or the overall system data rate. However, other quality of service (QoS) requirements, such as the strict deadlines of the M2M traffic, still require more attention with the deployment of drone base stations. Despite the flexibility of the UAV's mobility that is expected to provide considerable improvement to the wireless network capacity and coverage, the dynamic deployment of the UAV-aided communication systems has not received sufficient attention in the literature. This is due to the fact that the flexible mobility of UAVs increases the complexity of the formulated drone placement problem space. Such complexity led the previous research work to use simplifying assumptions by fixing one of the problem dimensions, such as the vertical or horizontal movement of drones, thus resulting in a static placement of the aerial deployment for the system model.

In this paper, we study the dynamic deployment of a single drone-mounted base station (BS) to provide LTE wireless communication coverage to a massive population of MTCDs. The deployed drone is to guarantee wireless connectivity for critical and strict communications of a group of MTCDs that are located out of the reach of a regular terrestrial infrastructure coverage. The main objective of this study is, therefore, to devise a joint optimal dynamic placement and uplink resource scheduling for the drone communication access while considering the diverse QoS requirements of the deployed M2M communications, especially for the delay-intolerant MTCDs.

1.1 Related Work

The study [4] utilizes multiple drones as communication relay nodes between different congested areas and a terrestrial BS. The deployed drones are responsible for serving small cells associated with the macro cell of the ground BS to improve the capacity of the overall network. Disaster situations and public safety applications are considered in [5] where drones are used to provide wireless communication coverage to the damaged area. The maximum achievable users' coverage is discussed in [6] considering the different QoS requirements of the served users. The QoS demands are modeled in terms of the channel signal-to-noise ratio (SNR), and the optimal drone placement is derived from multiple circular coverage disks of different SNR thresholds such that the maximum number of users is achieved by the intersections of these small cells. The extended work in [7] considers other important communication factors such as the transmission power. Furthermore, the users' mobility model is presented in [8] using the random walk model that simulates the movement direction by uniformly distributed angles between 0 and 2π while the movement speed is modeled by a random velocity between 0 and a maximum allowable speed. Consequently, the drone placement problem is solved in accordance with the modeled users' position changes. The drone positioning problem is also studied for indoor applications in [9] and [10] where the users are spread out over a high-rise building that is not covered by a terrestrial communication system. Basically, the drone placement problem is solved to minimize the total transmission power for uniformly distributed users in [9] and for symmetric user deployment in [10]. Further, the study [10] considers the worst served location in the building where the indoor users have the maximum path loss signal. The study in [11] assumes the presence of a terrestrial 3GPP base station for the deployed drones that function as small cells inside the supported ground-based cell in congestion relief applications. The major contribution of the study in [11] is to devise a 3D placement technique that clusters the served users with respect to their communication requirements under the consideration of the backhaul link allocation for the drone BS. In [12], a spectrum sharing technique is studied between a ground BS and a drone BS for load balancing applications in high traffic conditions. The optimization problem in this case jointly considers the drone's trajectory in a circular motion along the terrestrial BS cell edge as well as the physical resource scheduling and users clustering as edge and non-edge users such that the drone's altitude is constant and the users that are closer to the UAV have the highest priority to communicate via the drone. The study in [13] proposes a technique that optimizes the drone's altitude and the antenna beamwidth to maximize the overall system throughput. The users are then clustered into small cells, and each cell is served in turn by the drone BS in different communication modes. Another study, [14], considers the clustering technique for IoT devices to solve the drone placement problem along with allocating the available bandwidth. Multiple drones are employed in [14] to minimize the devices' transmission power while maintaining the achievable data rate. A fixed drone's trajectory is also used in [15] to solve the drone placement problem in serving a group of ground users. The optimization problem is concerned with the resource scheduling while the circular trajectory of the drone is exploited to enhance the communication links in a constant period of time. The M2M communications are considered in [16] where drones are utilized to gather information data from different clusters of MTCDs. Each cluster is served through a head node that collects the data from the clusters' devices. In [17], multiple drones are deployed to serve a group of users with an out-of-order terrestrial BS due to excessive traffic demands. Therefore, the optimization problem considers a system throughput-based cost function. The optimal drone's 3D position is obtained first for maximum coverage, then the backhaul link connectivity is ensured in advance of the resource scheduling.

In these previous studies, the formulated drone placement problem is solved to obtain a static drone's location or to follow a fixed trajectory for maximum geographical coverage with no guarantees of the communication coverage. Thus, this static drone deployment does not deal with the uncertainties of the traffic demands, especially in the diverse M2M communications. The formulated drone location optimization problems are basically modeled in these studies for an objective concerning the number of geographically covered users, the system throughput and the transmission power. Then, the optimization parameters are decoupled, and the formulated problems are solved in a sequence that develops conditional or special-case solutions relying on the first parameters obtained in the sequence, such as the drone's 3D positioning or the resource allocation. Furthermore, the network communication technologies are not taken into account in the drone BS problem models. Consequently, the QoS demands are modeled in terms of the channel path loss power which does not represent the critical and diverse QoS requirements of the

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deployed users, in particular the strict transmission deadlines in the M2M networks. In addition, the studies that consider the IoT applications are mainly employed the cluster-based network topologies. These clustering techniques are challenged due to the diverse nature of the MTCDs' QoS requirements that cannot be combined in one cluster within the actual distribution of the MTCDs.

Therefore, this study addresses the shortcomings of the previous work by developing a dynamic trajectory of a drone-mounted BS that uses the LTE technology for serving delay-sensitive M2M traffic. This trajectory depends on the MTCD traffic demands that vary with time. As shown in Fig. 1, the drone's mobility is utilized to respond to previously unknown M2M data traffic. This mobility improves the communication channels' connectivity which can lead to optimal resource allocation considering the strict transmission deadlines and the diverse QoS demands of the involved MTCDs. Therefore, the formulated problem is concurrently solved along the two axes that represent the drone's 3D placement and the network resource scheduling, respectively. This is done despite the complexity added to the problem due to the considerations of the UAV mobility and the QoS demands.

1.2 Paper Contributions and Organization

The contributions of this study can be summarized in the following:

- Optimizing the drone-mounted BS placement dynamically in such a way that maximizes the number of served MTCDs while minimizing the deadline missing ratio of each MTCD in the network. The resulting technique provides an exact solution but with high complexity. Hence, this solution serves as the optimal drone deployment bound.
- Devising an optimal resource allocation scheme jointly with the optimized drone deployment with concern to the diverse M2M traffic demands.
- Developing a lower-complexity optimization technique for the dynamic drone deployment problem to allow a real-time implementation of the drone-based communication system.

The rest of this paper is organized as follows. In Section 2, we introduce the system model and formulate the optimization problem. Section 3 introduces the exact optimization technique for the drone deployment problem. Then, the real-time heuristic technique is proposed in Section 4. Next, the modeled network with its associated parameter setup is presented in Section 5 along with the validation simulation experiments and results. Finally, the study is concluded in Section 6.

2 System Model and Problem Formulation

The system model considers a massive deployment of stationary MTCDs that are uniformly distributed over a space with a given density. Let $\mathbb{U}$ denote the set of deployed MTCDs characterized with different QoS requirements. The drone's location is denoted by $(x_d,\ y_d, H_d)$ while each MTCD i is located at $(x_i,\ y_i)$ within the distribution space. Thus, the direct distance between the drone BS and an MTCD i is calculated as

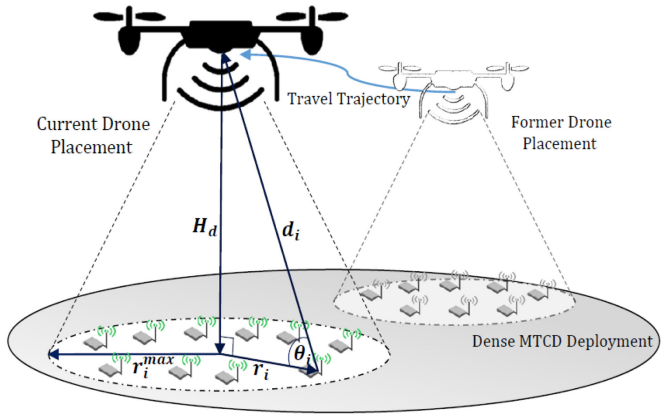

Fig. 1. Different drone deployment in response to MTCD's traffic demands.

$d_i^2 = {H_d}^2 + {r_i}^2$ where r_i is the ground distance between the MTCD's location and the drone's ground projection. Since we consider the drone BS to be equipped with an omnidirectional antenna, the associated drone's coverage is given in the form of a circular disk over the deployed space with a variable radius r_i^{max} according to the specified QoS metrics of each involved traffic type. The communication channels in this study are modeled in the uplink direction since the M2M traffic is mainly in the uplink links to transmit data to control stations for further processing. The air-to-ground links are modeled in terms of the path loss equations of both line-of-sight (LoS) and non-line-of-sight (NLoS) channels. These equations are given in [18] using the free-space path loss model as follows

$$L_{LoS} = 20\log\left(\frac{4\ \pi\ f_c d_i}{c}\right) + \zeta_{LoS}, \tag{1}$$

$$L_{NLoS} = 20\log\left(\frac{4\ \pi\ f_c d_i}{c}\right) + \zeta_{NLoS}, \tag{2}$$

where f_c is the carrier frequency, c is the speed of light and the additional losses in each spectral propagation scheme are expressed as ζ_{LoS} and ζ_{NLoS} , respectively. The probability of communicating over LoS/NLoS channel links is calculated, and the average path loss equation over both links can be expressed as

$$L_i = L_{LoS} P_{LoS} + L_{NLoS} P_{NLoS}, \tag{3}$$

where the probability of establishing a connection over LoS links, P_{LoS} , is given as

$$P_{LoS} = \frac{1}{1 + a\ exp(-b(\theta_i -\ a))}, \tag{4}$$

where a and b are environment-dependent parameters and θ_i is the elevation angle which is expressed as the arctangent of the drone height H_d over the base distance r_i of a right-angle triangle connecting the drone's 3D position with its corresponding ground projection and the location of an MTCD i. Since the propagation of the communication channels can either be LoS or NLoS at a time for each MTCD, the probability of establishing a connection over NLoS links can

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be derived from (4) as $P_{NLoS} = 1 - P_{LoS}$. The overall path loss equation of a communication channel between the drone and an MTCD i can, therefore, be calculated as

$$L_i = \frac{\zeta_{LoS} - \zeta_{NLoS}}{1 + a\exp(-b(\theta_i - a))} + 10\log(r_i \sec\theta_i) + 20\log\left(\frac{4\pi f_c}{c}\right) + \zeta_{NLoS}. \tag{5}$$

The channel model expressed in (5) is used to calculate the channel SNR value, given both the transmission and noise power, and determine the maximum drone coverage that allows all the MTCDs within the covered region to communicate over the transmission links specified with specific QoS requirements. Therefore, equation (5) is solved at different path loss thresholds L_i^{thr} that represent the minimum channel connectivity required to maintain the different QoS demands. Then, the maximum allowable elevation angle θ_i^{max} of the MTCDs at the drone cell radius r_i^{max} that corresponds to the pre-specified channel path loss can be driven from the gradient of equation (5) as

$$\frac{\pi}{9\ln 10}\tan\theta_i^{max} + \frac{ab(\zeta_{LoS} - \zeta_{NLoS})\ exp\big(-b\ (\theta_i^{max} - a)\big)}{\big(1 + a\ exp\big(-b\ (\theta_i^{max} - a)\big)\big)^2} = 0. \tag{6}$$

Subsequently, the corresponding cell radius r_i^{max} is obtained by solving the path loss model in (5) at θ_i^{max} and L_i^{thr} that guarantees maximum coverage with retained QoS requirements distributed among the covered MTCDs. This locates the drone BS at different height levels H_d^{max} based on the QoS required by each traffic profile. The maximum drone's height can be calculated for different traffic types as

$$H_d^{max} = r_i^{max}\tan\theta_i^{max}. \tag{7}$$

In LTE networks, the channel model given in (5) and its associated SNR value are incorporated into the channel quality indicator (CQI) that is reported to the BS by the served MTCDs and mapped to the modulation scheme to be used in data transmission [19]. Therefore, the achievable throughput per MTCD in kbits/sec can be calculated with respect to the decoded spectral efficiency η_i multiplied by the bandwidth BW_{RB} of the allocated physical resource blocks (PRBs) in one LTE transmission time interval (TTI) as follows

$$R_i = \eta_i BW_{RB} \sum\nolimits_{k\in\mathcal{N}_{RB}} I_{i,k}, \tag{8}$$

where $BW_{RB} = 180\ kHz$ [20], $I_{i,k}$ is a binary index that indicates whether the PRB k is assigned to the MTCD i and $\mathcal{N}_{RB}$ is a set of the total available PRBs according to the channel bandwidth, as given in [21].

In the LTE session management request message, the served MTCDs send to the BS information related to the way the data traffic should be handled using different indices such as the QoS class indicator (QCI) which implies the packet delay budgets DB_i [22]. The amount of data in the uplink buffer is communicated by the buffer status report (BSR) [23]. Since we consider in this study the various delay requirements and transmission deadlines of the deployed MTCDs that are assumed to generate Poisson traffic, the M/D/1 queue model is adjusted to estimate the total waiting time W_i of the data packets. The packet arrival rate denoted as λ_i is estimated by following a Poisson process to calculate the queue waiting time distribution as follows [24]

$$P[W_i \le t] = \left(1 - \frac{\lambda_i}{R_i}\right)\sum\nolimits_{\nu=0}^{z}\frac{[-\lambda_i(t - \nu T_i)]^\nu}{\nu!}exp(\lambda_i(t - \nu T_i)), \tag{9}$$

where T_i is a deterministic time in milliseconds required to transmit data packets pending in the uplink buffer and is equal to the inverse of the achievable data rate multiplied by the packet size reported through the BSR while z is an integer that satisfies the following condition $zT_i \le t \le (z+1)T_i$. Subsequently, the overall delay time is calculated as $D_i = W_i + T_i$, and by substituting D_i for W_i in (9), the probability distribution of the total transmission time combined with the waiting time can be calculated as [25]

$$P[D_i \le DB_i] = \left(1 - \frac{\lambda_i}{R_i}\right)\sum\nolimits_{\nu=0}^{z}\frac{[-\lambda_i(DB_i - T_i - \nu T_i)]^\nu}{\nu!}exp(\lambda_i(DB_i - T_i - \nu T_i)). \tag{10}$$

Then, the probability of missing a transmission assignment is given as $P[D_i > DB_i] = 1 - P[D_i \le DB_i]$ as the two events are mutually exclusive.

Since we study the dynamic placement of a drone BS serving a massive deployment of MTCDs, the optimization problem is devised to maximize the MTCDs' communication coverage considering the strict transmission deadlines of the M2M traffic. This can be achieved by optimizing the 3D location of the drone BS to enhance the communication channels that allow for optimal allocation of the available radio resources under the constraint of the MTCDs' deadlines. This optimization problem can, therefore, be written as

$$\underset{x_d,\, y_d,\, H_d, \boldsymbol{I}}{\text{maximize}} \quad \sum\nolimits_{i\in\mathbb{U}} u_i, \tag{11}$$

$$\textit{subject to} \quad H_d^{min} \le H_d \le H_d^{max}, \tag{11a}$$

$$P[D_i > DB_i] \le DM_i^{max}, \quad \forall i \in \mathbb{u}_a, \tag{11b}$$

$$\sum\nolimits_{i\in\mathbb{u}_a} I_{i,k} \le 1, \forall k \in \mathcal{N}_{RB}, \tag{11c}$$

$$-r_d \le x_d,\ y_d \le r_d, \tag{11d}$$

$$u_i \in \{0,1\}, \forall i \in \mathbb{U}, \tag{11e}$$

$$\boldsymbol{I} = [I_{i,k} \in \{0,1\}, \forall i \in \mathbb{u}_a,\ k \in \mathcal{N}_{RB}]^T, \tag{11f}$$

$$x_d,\ y_d,\ H_d \in \mathbb{R}, \tag{11g}$$

where $\mathbb{u}_a \subseteq \mathbb{U}$ is a subset containing the active MTCDs requesting wireless communication services, H_d^{min} is the lower bound of the drone's height while we consider the maximum allowable drone height H_d^{max} as calculated in (7) at an MTCD i with tolerant traffic requirements to allow for maximum drone coverage, DM_i^{max} is the upper bound given to the probability

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distribution of the transmission deadlines, r_d is the radius of the MTCD deployment space, $\boldsymbol{I}$ is a vector encompassing the resource allocation binary indicators and u_i is a piecewise function to indicate the geographical coverage of the deployed MTCDs by the drone BS. This function has to satisfy the constraint (11e) and is given as

$$u_i = \begin{cases} 1, & \sqrt{(x_i - x_d)^2 + (y_i - y_d)^2} \leq r_i^{max} \\ 0, & \sqrt{(x_i - x_d)^2 + (y_i - y_d)^2} > r_i^{max}. \end{cases} \tag{12}$$

Hence, both the drone's 3D placement and the scheduling of the PRBs are considered in the optimization problem in (11). This results in a mixed-integer nonlinear programming problem (MINLP) that is classified as an NP-hard problem. This class of NP-hard problems becomes intractable and needs exponential effort in the problem size that is linked to the large number of deployed MTCDs in our system model. In addition, the fitness function in (11) is non-differentiable because it has discontinuity jumps along the variable axes. This challenges the use of conventional optimization techniques that depend on gradient-based approaches either in finding the optimal solution or in linearizing the problem model. Moreover, the optimization problem in (11) is also found to be infeasible where the optimal solution cannot satisfy some of the problem constraints, such as the constraint in (11b). This constraint cannot be fulfilled for all active MTCDs sending scheduling requests since the uplink channels become occupied with a high number of the required communications. Based on the above and since the continuous variables concerning the 3D placement problem are coupled with the binary variables responsible for the resource allocation, the optimization problem becomes nonconvex in the feasible domain.

3 Proposed UAV Optimal Deployment Technique

We first propose a technique that provides the optimal bound of the solution to the problem in (11). For this purpose, we solve the optimization problem in (11) using both the penalty function method and the outer-approximation technique, combined, as proposed in [26]. The idea behind this approach is that the original MINLP problem can alternatively be solved in a series of relaxed nonlinear programming (NLP) subproblems and linearly approximated mixed-integer linear programming (MILP) master problems. The outer-approximation method allows to represent the nonlinearities of the continuous feasible space in a linear form that reduces the complexity of solving the original MINLP problem in (11) that may need to be solved for all possible permutations in the worst case. In addition to the outer-approximation approach, the proposed algorithm exploits the penalty function method to introduce some augmented variables to the relaxed MILP master problem to deal with the nonconvexity and the infeasibility of the original problem.

3.1 Solution Details

According to the procedures proposed in [27], we first estimate a starting point close to the optimal solution to avoid getting trapped in a local optimum point. Therefore, some heuristics are added to the algorithm to start at a point that can boost the exact optimal analysis performance. At the initial point, we fix the binary variables and solve the resulting NLP subproblem to find the optimum continuous variables that represent the drone's 3D placement at which we generate the linearized model of the original problem in (11). For this purpose, we use the Nelder-Mead algorithm [28] to solve the NLP subproblem at hand. This optimization technique does not use any differentiation records to search for the optimal point but keeps changing the worst vertex of a simplex connecting potential points over the search space. Hence, we use the penalty method [29] to formulate the subproblem in the form of an unconstrained optimization problem. Some penalty terms are introduced to the objective function for the NLP subproblem constraints. This method is tolerant of the infeasible constraint functions given in the original problem, so the derived NLP subproblem is stated as

$$\begin{aligned} \underset{x_d, y_d, H_d}{\text{maximize}}\ & f(x_d,\ y_d, H_d) = \sum_{i \in \mathbb{U}} u_i - \sum_{i \in \mathbb{u}_a} \psi_i \\ & C_i(x_d,\ y_d, H_d),\ \psi_i > 0,\ \forall i \in \mathbb{u}_a, \end{aligned} \tag{13}$$

where ψ_i is the penalty coefficient that controls each constraint's penalties denoted by $C_i(x_d,\ y_d, H_d)$. These exact penalty functions are limited to the number of the constraint functions stated in (11) and are given as follows

$$\begin{aligned} C_i = & \max\big(0, H_d - H_d^{max}, H_d^{min} - H_d\big)^2 \\ & + \max\big(0, P[D_i > DB_i] - DM_i^{max}\big)^2,\ \ \forall i \in \mathbb{u}_a. \end{aligned} \tag{14}$$

The rest of the constraint functions in (11), such as the constraint in (11c) that ensures a single allocation of all available PRBs to the MTCDs sending data packets, are to be satisfied while estimating the starting point of the search algorithm, which specifies the starting resource allocation scheme. This starting point will be kept unchanged in the iterations of the NLP solver. On the other hand, the upper and lower bounds of the continuous variables representing the 2D coordinates of the drone's location are considered during the initial construction of the simplex vertices in the Nelder-Mead algorithm. The vertices update process guarantees that the simplex would always be within the limits of the variables as it tries to explore all potential points that are certainly bounded by the deployed ground space.

By solving the NLP subproblem, the obtained solution $\boldsymbol{V}_0$ that represents a potential 3D placement (x_d^0, y_d^0, H_d^0) of the drone BS for the given PRBs allocation is used to build an approximate linear model of the original problem in (11). Hence, the MILP master problem is attained using the outer-approximation method around the potential solution $\boldsymbol{V}_0$ as

$$\underset{\boldsymbol{V},\ \boldsymbol{I},\ \mu}{\text{maximize}} \quad \mu, \tag{15}$$

$$\textit{subject to}\ \ \mu \leq g(\boldsymbol{V}_0) + \nabla g(\boldsymbol{V}_0)^T(\boldsymbol{V} - \boldsymbol{V}_0), \tag{15a}$$

$$h(\boldsymbol{V}, \boldsymbol{I}) \leq 0, \tag{15b}$$

$$\sum\nolimits_{i \in \mathbb{u}_a} I_{i,k} \leq 1,\ \forall k \in \mathcal{N}_{RB}, \tag{15c}$$

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$$\boldsymbol{V}^{Lb} \leq \boldsymbol{V} \leq \boldsymbol{V}^{Ub}, \tag{15d}$$

$$\boldsymbol{I} = [I_{i,k} \in \{0,1\}, \forall i \in \mathbb{u}_a,\ k \in \mathcal{N}_{RB}]^T, \tag{15e}$$

$$\mu \in \mathbb{R}\,,\ \boldsymbol{V} \in \mathbb{R}^3, \tag{15f}$$

where μ is a variable introduced and bounded by the equivalent linear objective function and $g(\boldsymbol{V})$ is the fitness function of the original problem in (11) and calculated in (12) by having a vector $\boldsymbol{V}$ of the continuous variables x_d, y_d and H_d while $h(\boldsymbol{V}, \boldsymbol{I})$ is the probability function of missing a transmission assignment which is stated as a constraint in the original problem in (11b) and given as the compliment of Equation (10). The rest of the constraints are kept unchanged as they are linearly stated in (11), where the lower and upper bounds of the continuous variables are stored in $\boldsymbol{V}^{Lb}$ and $\boldsymbol{V}^{Ub}$, respectively. Since the nonlinearities are embedded in the problem variables in the constraint (15b) concerning the probability of deadline missing, this control limit is to be reformulated in terms of the minimum required data rate R_i^{min} that maintains the probability of deadline missing lower than the pre-specified probability limit. This change is required to segregate the integer and continuous variables in the linearized model. Hence, this constraint is rewritten as follows

$$\sum\nolimits_{k \in \mathcal{N}_{RB}} q_i(\boldsymbol{V}) I_{i,k} \geq R_i^{min}, \forall i \in \mathbb{u}_a, \tag{16}$$

where $q_i(\boldsymbol{V})$ is a function of the drone's 3D location that is responsible for estimating the communication links' CQI index for each MTCD i and the corresponding channel spectral efficiency η_i which controls the achievable data rate, as given in (8). In order to linearize the above formulation, other constraints have to be introduced to the linear model in (15), and the summation is to be separated in the following constraints by defining $R_{i,k}$ that denotes the achievable data rate by assigning a single PRB k to a single active MTCD i.

$$\sum\nolimits_{k \in \mathcal{N}_{RB}} R_{i,k} \geq R_i^{min}, \forall i \in \mathbb{u}_a, \tag{17}$$

$$R_{i,k} \leq R^{max} I_{i,k}, \forall i \in \mathbb{u}_a,\ k \in \mathcal{N}_{RB}, \tag{18}$$

$$R_{i,k} \geq 0, \forall i \in \mathbb{u}_a,\ k \in \mathcal{N}_{RB}, \tag{19}$$

$$R_{i,k} \leq q_i(\boldsymbol{V}_0) + \nabla q_i(\boldsymbol{V}_0)^T (\boldsymbol{V} - \boldsymbol{V}_0), \forall i \in \mathbb{u}_a,\ k \in \mathcal{N}_{RB}, \tag{20}$$

$$R_{i,k} \geq q_i(\boldsymbol{V}_0) + \nabla q_i(\boldsymbol{V}_0)^T (\boldsymbol{V} - \boldsymbol{V}_0) + R^{max} I_{i,k} - R^{max}, \quad \forall i \in \mathbb{u}_a,\ k \in \mathcal{N}_{RB}, \tag{21}$$

where R^{max} is the maximum achievable throughput in the LTE communication channels.

Moreover, an augmented penalty variable ρ is introduced to the objective function of the master problem in (15) as proposed in [26] since the assumption of convexity cannot be met by the optimization model in (11) and the conventional outer-approximation technique does not guarantee convergence to an optimum global point unless the original problem is convex. Furthermore, other penalty variables σ_i are to be added to the fitness function in (15) since the model is infeasible to satisfy the constraint in (16) because of the scarce communication resources given the large number of active MTCDs. Hence, the MILP master problem can be restated as

$$\underset{\boldsymbol{V},\, \boldsymbol{I},\, \mu, \boldsymbol{R}, \rho,\, \sigma_i}{\text{maximize}} \quad \mu - \omega\rho - \sum\nolimits_{i \in \mathbb{u}_a} \omega_i \sigma_i, \tag{22}$$

$$\textit{subject to} \quad \mu \leq g(\boldsymbol{V}_0) + \nabla g(\boldsymbol{V}_0)^T (\boldsymbol{V} - \boldsymbol{V}_0), \tag{22a}$$

$$R_{i,k} \leq R^{max} I_{i,k},\ \forall i \in \mathbb{u}_a,\ k \in \mathcal{N}_{RB}, \tag{22b}$$

$$R_{i,k} \leq q_i(\boldsymbol{V}_0) + \nabla q_i(\boldsymbol{V}_0)^T (\boldsymbol{V} - \boldsymbol{V}_0), \quad \forall i \in \mathbb{u}_a,\ k \in \mathcal{N}_{RB}, \tag{22c}$$

$$R_{i,k} \geq q_i(\boldsymbol{V}_0) + \nabla q_i(\boldsymbol{V}_0)^T (\boldsymbol{V} - \boldsymbol{V}_0) + R^{max} I_{i,k} - R^{max}, \forall i \in \mathbb{u}_a,\ k \in \mathcal{N}_{RB}, \tag{22d}$$

$$R_{i,k} \geq 0,\ \forall i \in \mathbb{u}_a,\ k \in \mathcal{N}_{RB}, \tag{22e}$$

$$\sum\nolimits_{k \in \mathcal{N}_{RB}} R_{i,k} + \omega_i \sigma_i \geq R_i^{min},\ \forall i \in \mathbb{u}_a, \tag{22f}$$

$$\sum\nolimits_{i \in \mathbb{u}_a} I_{i,k} \leq 1,\ \forall k \in \mathcal{N}_{RB}, \tag{22g}$$

$$\boldsymbol{V}^{Lb} \leq \boldsymbol{V} \leq \boldsymbol{V}^{Ub}, \tag{22h}$$

$$\boldsymbol{I} = [I_{i,k} \in \{0,1\},\ \forall i \in \mathbb{u}_a,\ k \in \mathcal{N}_{RB}]^T, \tag{22i}$$

$$\boldsymbol{R} = [R_{i,k} \in\ \mathbb{R}\,, \forall i \in \mathbb{u}_a,\ k \in \mathcal{N}_{RB}]^T, \tag{22j}$$

$$\mu,\ \omega,\ \rho,\ \omega_i,\ \sigma_i \in \mathbb{R}, \forall i \in \mathbb{u}_a, \boldsymbol{V} \in \mathbb{R}^3, \tag{22k}$$

where ω is the penalty coefficient and ω_i is a variable weight given to the penalty variables σ_i that allows the delay-intolerant traffic to have a transmission priority over the other traffic types. Therefore, these weights are bounded as follows

$$\omega > 0, \omega_i < 0, \forall i \in \mathbb{u}_a. \tag{23}$$

Since the master problem in (22) is not differentiable, the multiple linear regression modeling method [30] is used in linearizing the objective function $g(\boldsymbol{V})$ and the constraint function $q_i(\boldsymbol{V})$ instead of the conventional differentiation-based linearization technique. Thus, the assumptions of multiple linear regression are to be considered. First, the objective function $g(\boldsymbol{V})$ and the constraint function $q_i(\boldsymbol{V})$ are required to have a linear relationship between the outcomes and the independent variables. In this problem model, the number of covered users is directly proportional to the drone's height while it is slightly changing by moving the drone horizontally because the users are uniformly distributed with a constant density over the deployed space. On the other hand, increasing the drone's altitude inversely affects the communication channel quality. Consequently, the assumption of linearity can be met by the functions model, especially when we linearize the functions $g(\boldsymbol{V})$ and $q_i(\boldsymbol{V})$ just around the potential point obtained by solving the NLP subproblem. Furthermore, the normality of the residuals is to be satisfied since the

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linearization model is formed by observations fairly close to the solution of the NLP subproblem. Also, we assume that the variables are independent and not correlated as they represent the 3D coordinates of the drone BS. Then, the residuals do not depend on each other, and the variance of the residuals is kept persistent through the fitted regression line. As a result, the derived MILP master problem in (22) can render an adequate linear model replacing the original MINLP problem in (11). The master problem in (22) is solved using the branch-and-cut algorithm [31]. This exact optimization method incorporates the cutting-plane technique to the conventional branch-and-bound algorithm.

To find the optimal solution of the original problem in (11), the optimization algorithm iterates to explore potential points by maintaining the integer solution of the MILP master problem and solving the resulting NLP subproblem. The solutions of the relaxed NLP subproblems are to define confined constraints to the feasible space of the original problem (11). These constraints introduce the branching rules of the branch-and-bound method, as proposed in [27]. However, suppose the linear solution does not provide further improvement to the original problem. In that case, the algorithm starts to estimate other integer combinations of the binary variables derived from the best-obtained solution in order to provide a broader exploration of the search space. Algorithm 1 demonstrates the sequence of the procedures of the exact optimization technique.

3.2 Computational Complexity of the Optimal Solution

The computational complexity of the proposed exact optimization technique in Algorithm 1 can be analyzed and presented in the form of the big-O notation. To this end, we state the worst-case complexity of the computational steps given in Algorithm 1 as follows.

- Steps 1&3 require constant time complexity $O(1)$.
- Step 2 contributes $O(1) + O(nt) + O(t) + O(nt\log(n)) + O(t\log(n))$.
- Step 5 needs $O(Cn^2t) + O(n^3t) + O(nt) + O(t)$.
- Step 6 needs exponential time complexity $O(C^n t)$.
- Steps 7-27 contribute $O(t) + O(nt^2) + O(t^2) + O(nt^2\log(n)) + O(t^2\log(n))$.

In the aforementioned operations computational complexity, n denotes the algorithms' input variables size that combines the drone's 3D coordinates and the available network resources for all deployed MTCDs in solving the master problem while for the subproblem solution, the size of the problem dimensions n is equal to the size of the drone placement vector V augmented by one. Also, C is a constant greater than one dependent on the solved MILP optimization problem feasible space and dimensions. Moreover, the overall algorithm contains an outer loop of iterations t and an inner loop with the size of the original problem dimensions n. Therefore, the total computational complexity of the proposed optimal solution is the summation of each algorithmic step's complexity. However, by considering the fastest growing term of the computational operations, the overall time complexity of the exact algorithm becomes exponential as a result of solving the MILP master problem.

Algorithm 1: Exact optimal solution for the drone BS deployment

Input: $f(\boldsymbol{V})$, $g(\boldsymbol{V})$, $q_i(\boldsymbol{V})$, $\boldsymbol{V}^{Ub}$, $\boldsymbol{V}^{Lb}$, $MaxIter$
Output: $V^* : V^* = [x_d^* \; y_d^* \; H_d^*]^T$, $\boldsymbol{I}^* : \boldsymbol{I}^* = [I_{i,k}^*, \forall i \in \mathbb{u}_a, \; k \in \mathcal{N}_{RB}]^T$

1. estimate an initial or starting search point using some heuristics
2. fix the integer vector $\boldsymbol{I}_0^{init}$ of the estimated initial point, then solve the resultant NLP subproblem and obtain $\boldsymbol{V}_0^{init}$
3. calculate the objective value Z^{init} at $(\boldsymbol{V}_0^{init}, \boldsymbol{I}_0^{init})$ and set the lower bound $Z^{Lb} = Z^{init}$, $\boldsymbol{V}_0 = \boldsymbol{V}_0^{init}$, $\boldsymbol{I}_0 = \boldsymbol{I}_0^{init}$, $cnt = 0$ // counter reset
4. **for** $t \leftarrow MaxIter$ // maximum number of algorithm iterations
5. linearize the master problem around the solution of the NLP subproblem $\boldsymbol{V}_0^t$ using outer approximation and multiple linear regression techniques // $\boldsymbol{V}_0^t = \boldsymbol{V}_0^{init}$ at the first iteration
6. set up the MILP master problem in (22) and solve the linear model using the branch-and-cut method
7. **if** the MILP master problem solution is infeasible, **then**
8. let $\boldsymbol{I}_0^t = \boldsymbol{I}^{Lb}$ // $\boldsymbol{I}_0^t$ is a vector of the integer solution of the MILP master problem in iteration t
9. update $\boldsymbol{I}_0^t$ by changing one integer variable state
10. **else**
11. accept the integer solution $\boldsymbol{I}_0^t$
12. **end if**
13. solve the NLP subproblem at the integer vector $\boldsymbol{I}_0^t$
14. calculate Z^t at $\boldsymbol{V}_0^t$ and $\boldsymbol{I}_0^t$ // Z^t is the objective value in iteration t
15. **if** $Z^t > Z^{Lb}$, **then**
16. update $Z^{Lb} = Z^t$, $\boldsymbol{V}_0 = \boldsymbol{V}_0^t$, $\boldsymbol{I}_0 = \boldsymbol{I}_0^t$
17. reset cnt, continue
18. **else**
19. increment cnt
20. **if** $cnt \leq 0.25 \times MaxIter$, **then**
21. update $\boldsymbol{I}_0^t$ by changing one integer variable state
22. go to step 13
23. **else**
24. break
25. **end if**
26. **end if**
27. **end**
28. $\boldsymbol{V}^* = \boldsymbol{V}_0$, $\boldsymbol{I}^* = \boldsymbol{I}_0$

4 Proposed UAV Heuristic Deployment Technique

Since the optimal solution of the optimization problem in (11) is intractable and computationally expensive when employing exact optimization techniques, we propose a low-complexity heuristic-based technique for the drone deployment problem for the use in real-time arrangements. For this purpose, we use the particle swarm optimization (PSO) algorithm [32] to provide a near-optimal real-time solution to the problem in (11). It has been shown that PSO outperforms other meta-heuristic approaches with its fast convergence and wide exploration of high-dimensional decision spaces with minimum computational effort [33]. Moreover, PSO provides a suitable dynamic architecture in search of potential solutions in the formulated problem discontinuity of the feasible region when compared with other

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meta-heuristic and evolutionary approaches such as the conventional genetic algorithm (GA) that uses mutation and crossover processes in updating potential points. These GA procedures do not provide an inclusive search in the high-dimensional feasible set of the formulated problem. In a problem with a large number of variables given in [34], the PSO-based technique is found to be superior compared to other meta-heuristic optimization approaches in terms of solution precision, robustness and convergence speed.

4.1 Solution Details

To proceed with the use of the PSO-based solution, the drone deployment problem is to be reformulated and represented as an unconstrained objective function. This procedure is done by the penalty method [29] by adding some constraint penalty functions to the objective function in (11). This allows the search algorithm to be tolerant of the infeasible constraints due to the system limits in serving the large number of MTCDs requesting communication access. The derived unconstrained objective function is given similar to the one in (13) but in all the variables representing the drone's 3D coordinates and the resource scheduling to the active MTCDs. The unconstrained optimization problem can be reformulated from the original problem in (11) as follows

$$\begin{aligned}
&\underset{\boldsymbol{V},\,\boldsymbol{I}}{\text{maximize}}\, f(\boldsymbol{V},\ \boldsymbol{I}) = \sum\nolimits_{i\in\mathbb{U}} u_i \\
&\quad - \sum\nolimits_{i\in\mathbb{u}_a} \psi_i \max\left(0, P[D_i\ >\ DB_i] - DM_i^{max}\right)^2,\ \psi_i\ >\ 0, \\
&\quad u_i \in \{0,1\},\ \forall i \in \mathbb{U},\ \boldsymbol{V} \in \mathbb{R}^3, \\
&\quad \boldsymbol{I} = [I_{i,k} \in \{0,1\},\ \forall i \in \mathbb{u}_a,\ k \in \mathcal{N}_{RB}]^T.
\end{aligned} \tag{24}$$

The other constraints in (11a), (11c) and (11d) are considered in the initialization process of a particle swarm set, denoted by $\mathcal{S}$, and enforced while modifying the particles' locations through their updated velocities and directions. Furthermore, the penalty coefficient ψ_i dynamically changes according to the traffic demands giving high priority to the strict data transmissions. Since the derived optimization problem in (24) contains some binary variables, the velocity update process of the swarm particles is to be reconsidered as it depends on measuring the distances between particles' best locations. These distances are continuous in the problem domain. As a result, we integrate the discrete form of PSO proposed in [33] to the near-optimal solver. In the algorithm implementation, an alternative distance measurement is defined for the binary variables. This integer-reformulated continuous dimension is denoted as the probability of changing a binary state. This state transition probability is mapped to the potential resource scheduling scheme that is used in calculating the unconstrained objective function in (24). Therefore, the problem in (24) is solved considering both the binary and continuous variables in the particles search procedures using two different velocity update concepts. Moreover, we utilize the unified model of PSO given in [35] to ensure convergence to a near-global optimum and to avoid the particles getting trapped in local points since they are allowed to learn from personal, local and global exemplars at each iteration of the algorithm. The detailed procedures of the proposed heuristic technique are illustrated in Algorithm 2.

Algorithm 2: PSO-based solution for the drone BS deployment

Input: $f(\boldsymbol{V},\ \boldsymbol{I})$, ψ_i, $\boldsymbol{V}^{Ub}$, $\boldsymbol{V}^{Lb}$
Output: $V^* : V^* = [x_d^*\ y_d^*\ H_d^*]^T$, $\boldsymbol{I}^* : \boldsymbol{I}^* = [I_{i,k}^*, \forall i \in \mathbb{u}_a,\ k \in N_{RB}]^T$

1. Initialize uniformly distributed swarm $\mathcal{S}$ of particles $\boldsymbol{P}_s$ of size $s_z : s_z = |\mathcal{S}|$ over the problem space such that $\boldsymbol{P}_s = [\boldsymbol{V};\ \boldsymbol{I}]_s$ (//vertical vector concatenate) within the variables upper/lower bounds $\boldsymbol{V}^{Ub}$, $\boldsymbol{V}^{Lb}$, $I_{i,k}^P \in [0,1]$, $\forall i \in \mathbb{u}_a,\ k \in N_{RB}$ //$I_{i,k}^P$ is the probability representation of the binary variables in $\boldsymbol{I}$
2. set PSO parameters $s_z, \phi, w, c_1, c_2,\ rg_z\ , MaxIter$ //swarm size s_z, unification factor ϕ, inertia constant w, acceleration constants c_1 and c_2, neighbors ring size rg_z, maximum iterations $MaxIter$
3. $stagnant\ cnt = 0$, $\mathrm{r}f = 0.05 \times MaxIter$ //stagnation detection count, refresh gap limit
4. calculate $f(\boldsymbol{P}_s\ )$ at all swarm particles //after mapping the probability status $I_{i,k}^P$. to $I_{i,k} \in \{0,1\}\ \forall i \in \mathbb{u}_a,\ k \in N_{RB}$
5. sort $\boldsymbol{F}$ the resulting vector of the objective function computations
6. determine global best particle $\boldsymbol{Gb}$ and local best particle $\boldsymbol{Lb}_s$ for all $s \in \mathcal{S}$ between each particle's neighbors in a ring of size rg_z
7. initialize particles' velocities $\boldsymbol{V}_s = rand(s_z, |\boldsymbol{P}_s|), \forall s \in \mathcal{S}$
8. **for** $t\ \leftarrow MaxIter$
9. **if** $cnt\ >\ rf$, **then**
10.   **if** $stagnant\ cnt == 0$, **then**
11.   set $stagnant\ cnt$ and reset cnt
12.   reinitialize $\boldsymbol{V}_s = rand(s_z, size\ of\ particle), \forall s \in \mathcal{S}$
13.   **else** terminate
14.   **end if**
15. **end if**
16. obtain updated particles' locations such that $\boldsymbol{P}_s\ : \boldsymbol{P}_s(1:3) \in [\boldsymbol{V}^{Ub},\ \boldsymbol{V}^{Lb}]$ // only for the continuous variables
    $\boldsymbol{P}_s\ : \boldsymbol{P}_s(4:end) \in [0,\ 1]$ //only for the binary variables
17. compute $f(\boldsymbol{P}_s)$ at all swarm particles, sort the resulting vector $\boldsymbol{F}$
18. determine updated $\boldsymbol{Gb}$, $\boldsymbol{Lb_s}$ and the personal best location $\boldsymbol{Pb}_s, \forall s \in \mathcal{S}$
19. calculate the updated w as per [35]
20. update particles' velocities $\boldsymbol{V}_s, \forall s \in \mathcal{S}$
21. normalize particles' velocities $\boldsymbol{V}_s(4:end)$ for the binary variables, $\forall s \in \mathcal{S}$
22. **if** no change in the $\boldsymbol{Gb}$ location, **then**
23.   increment cnt
24. **else**
25.   reset $stagnant\ cnt, cnt$
26. **end if**
27. **end**
28. $V^* = \boldsymbol{Gb}(1:3)$, $\boldsymbol{I}^* = \boldsymbol{Gb}(4:end)$

4.2 Computational Complexity of the Heuristic Solution

The proposed heuristic-based technique is expected to be relatively inexpensive in computations and able to perform in real-time arrangements. We present the complexity of the algorithm steps in terms of the big-O notation as follows.

- Steps 1&4 require linear time complexity $O(s_z)$.
- Steps 2&3 require constant time complexity $O(1)$.

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- Step 5 contributes $O(s_z \log(s_z))$.
- Step 6 contributes $O(\log(s_z)) + O(s_z \log(rg_z))$.
- Steps 7-27 need $O(t) + O(s_z t) + O(t \log(s_z)) + O(s_z t \log(rg_z)) + O(s_z t \log(s_z))$.

As indicated in the PSO algorithm steps complexity, the computational operations are mainly dependent on the swarm population size $s_z : s_z = |\mathcal{S}|$ that is relatively much smaller than the original problem dimensions. Furthermore, the total number of algorithm iterations t can be decreased by introducing a tolerant termination criterion. Hence, the remaining computational effort is encountered by the objective function calculations in each particle's location and by determining the personal, local and global best performance of each particle. To determine the overall time complexity of the proposed heuristic solution, the worst-case complexity of the PSO algorithm can be stated in $O(s_z^2 t)$ by considering the fastest growing terms in the algorithm execution that is mainly represented in the "sort-and-find" procedures. This overall worst-case time complexity is quadratic in the swarm size s_z while it is linear in the algorithm outer loop t. Since the complexity of this heuristic-based approach is independent of the high dimensions of the formulated optimization problem n, the heuristic solution's complexity is clearly much lower than that of the optimal solution.

5 Experimental Validation

5.1 Network Setup

The modeled drone-associated LTE network is simulated in a MATLAB environment subject to the parameters given in Table 1. The deployed population of stationary MTCDs is assumed to belong to a monitoring application that is deployed over a remote suburban area in the absence of a regular cellular infrastructure. Furthermore, we assume that the deployed MTCDs are uniformly distributed with a fixed density. Thus, by increasing the number of the deployed devices, the required drone's service area increases. The M2M traffic modeled in this study includes three different wireless device types, as shown in Table 2. The first traffic profile represents alarm MTCDs that transmit small data packets with strict deadlines. The other two types are delay-tolerant traffic profiles represented by wireless cameras and monitoring sensors. The camera traffic requires a high data rate to be able to transmit large data packets of images or videos. As such, the minimum data rate required to maintain both the delay and throughput requirements is obtained using the allowable probability of missing a transmission assignment given in Table 1. The delay bounds of each traffic profile are modeled in a uniform distribution within a delay span of the bounds indicated in Table 2. The dynamic deployment of the drone BS updates the communication link capacity according to the free-space path loss model and the current drone's position. The MTCDs' transmission and noise power specified in Table 1 are used to determine the channel SNR given the estimated channel attenuation.

TABLE 1
Network and Algorithms Parameters

Parameter	Value
Channel Bandwidth	3 MHz
Number of PRBs	15
Carrier Frequency f_c	2 GHz
Number of MTCDs	50, 75, 100, 125, 150
MTCDs Distribution Density	11 MTCDs/km^2
Environment Parameters a, b, ζ_{LoS}, ζ_{NLoS}	5.0188, 0.3511, 0.1, 21 [18]
Transmission, Noise Power	30, -70 dBm [6]
Deadline Missing Probability DM_i^{max}	10%
Nelder-Mead Algorithm Constants Shrink, Reflection, Contraction, Expansion	0.5, 1, 0.5, 2
Number of Regression Observations	100
PSO Parameters s, α, w, c_1, c_2, rg	50, 0.1, 0.729, 1.494, 1.494, 5 [35]
Algorithms Maximum Iterations $MaxIter$	250

5.2 Experimental Results

The optimal solution is obtained by generating the modeled M2M traffic and solving the formulated optimization problem using Algorithm 1. Then, the same network scenarios are solved using the heuristic approach of Algorithm 2. The parameters and constants used in these algorithms are given in Table 1. The obtained results are the averages of 10 independent simulation runs with 95 percent confidence intervals indicated for the different results. Each simulation run lasts for a duration of 1000 consecutive TTIs.

The results of the exact optimization technique are compared with those of the near-optimal solution obtained by the heuristic algorithm. The results of the two algorithms are also compared to those of the conventional stationary solution where the drone BS is deployed at a fixed height at the cell center. This stationary deployment places the drone at the optimal heights obtained in [18] where the resulting channel conditions are maintained within a threshold path loss value. We also integrate the obtained deployment in [18] with the delay-based scheduler in [36] (we will call this integrated system "the conventional solution" in the rest of this discussion) to evaluate the communication coverage of the proposed techniques. The system deadline missing ratio along with the system aggregate throughput are used as evaluation metrics of the overall network performance.

Fig. 2 illustrates the overall system deadline missing performance of the three techniques as we change the number of the deployed MTCDs. Both the optimal and heuristic solutions clearly outperform the conventional solution. For the small deployments of up to 75 MTCDs, the deadline missing does not exceed 10 percent of the total data transmissions. The alarm deadline missing ratio shown in Fig. 3 has approximately an equal percentage among the three approaches for small deployments of up to 75 MTCDs. The reason behind

TABLE 2
Traffic Configurations

Traffic Description	Alerts	Camera	Monitoring Sensor
Arrival Rate (pkt/s)	25	30	1
Packet Size (byte)	32	512	128
Profile Percentage %	20	20	60
Threshold Path Loss (dB)	95	98	100
Delay Budget (ms)	U(10,20)	U(125,250)	U(800,900)

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these performance results is that the delay-based scheduler is biased towards the traffic characterized by strict deadlines. Therefore, it assigns all the available PRBs to this type of traffic only regardless of other M2M traffic demands. In addition, the small deployment of MTCDs helps ensure sufficient channel conditions for all network devices even with the static drone positioning, so these small deployments assist the delay-based scheduler to serve the alarm traffic properly. However, the deadline missing occurrences of the camera traffic shown in Fig. 4 increases dramatically even with the small deployments in case of the stationary placement of the drone BS. On the other hand, the dynamic placement of the aerial system derived by both the optimal and heuristic solutions assists the configured resource allocation scheme for better utilization of the available PRBs through the enhanced channel conditions for a large number of active MTCDs. The proposed techniques, therefore, succeed in reducing the probability of missing the transmission deadlines by achieving a deadline missing ratio of less than 50 percent in all the modeled M2M deployments, especially in the larger deployments of 125 and 150 MTCDs, as shown in Figs. 3 and 4.

In terms of the system aggregate throughput, as illustrated in Fig. 5, the proposed optimal solution along with the heuristic solution outperform the conventional technique by obtaining higher throughput values, compared to the conventional solution, through the optimal utilization of the available channel bandwidth and its associated PRBs. The system throughput for the delay-tolerant traffic profiles, as shown in Figs. 6 and 7, declines significantly approaching zero data rate, in the case of the conventional solution, for the sensor traffic with the larger M2M deployments. On the other hand, the drone's mobility is utilized in the proposed techniques to maintain the channel attenuation at the required acceptable values that would guarantee uninterrupted communications for both the strict and delay-tolerant users.

Moreover, we assess the best drone placement derived by the proposed optimal solutions by calculating the mean average of the SNR value of the modeled LTE network. This SNR metric is used for the technique of the optimal static deployment derived in [6] to be compared with the dynamic drone placement obtained by the proposed techniques, see Fig. 8. The mean SNR indicates that the dynamic deployment of the drone BS maintains the channel conditions of the M2M network

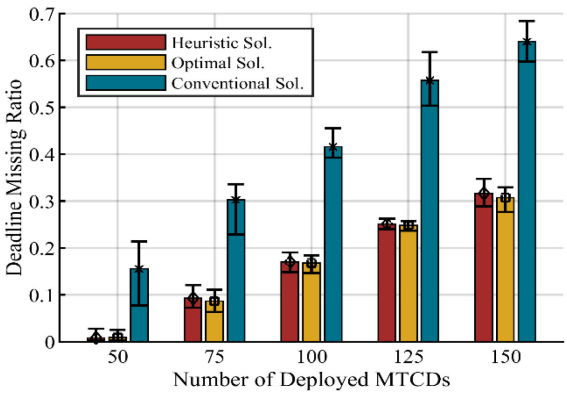

Fig. 2. Overall system deadline missing ratio.

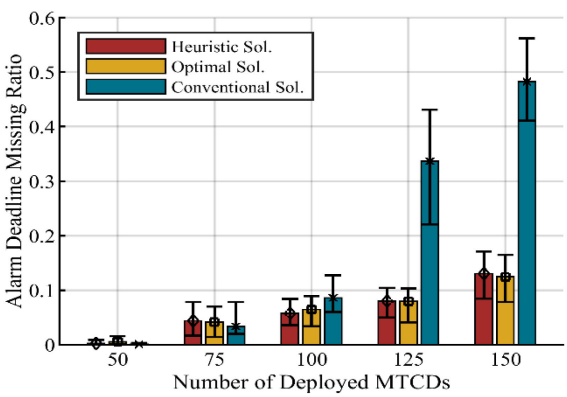

Fig. 3. Alarm traffic deadline missing ratio.

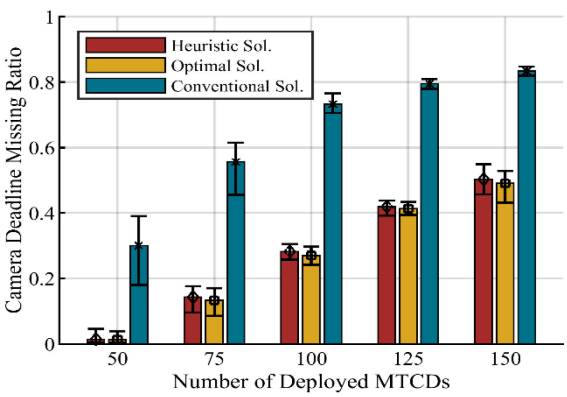

Fig. 4. Camera traffic deadline missing ratio.

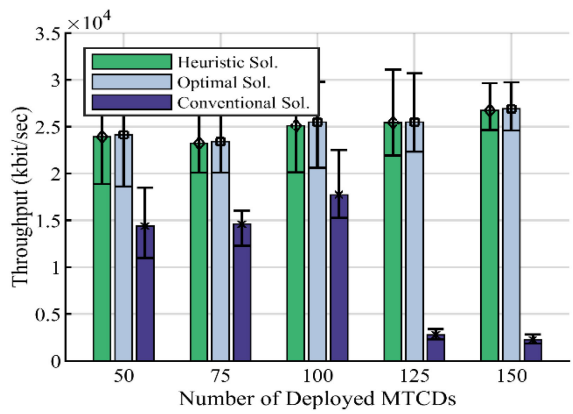

Fig. 5. Overall system throughput.

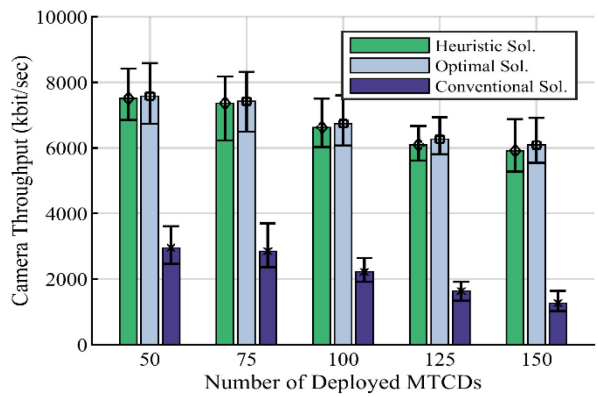

Fig. 6. Camera traffic data rate.

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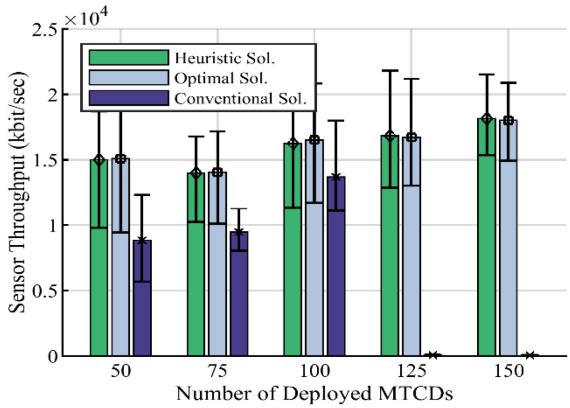

Fig. 7. Monitoring sensors data rate.

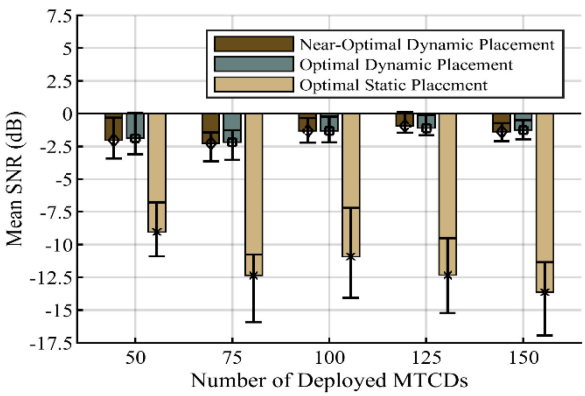

Fig. 8. Overall network SNR average.

devices within a value not exceeding -2.3 dB. These are the conditions required to attain the minimum value of the achievable data rates that can be acquired by the drone BS to meet the delay requirements of the deployed MTCDs. This SNR value is further increased by deploying a large number of MTCDs with the dynamic drone placement. The reason behind this performance is that the controlled drone's mobility is exploited to provide communication coverage for only the active MTCDs that are sending scheduling requests at that moment in time. Moreover, the deployed network becomes broader in the massive MTCD deployment where the stationary deployment of the drone BS cannot cover all the devices with the channel conditions required to fulfill the traffic QoS demands.

Since the heuristic technique produces results that are closely comparable to those of the optimal solution, this makes it a strong candidate as a real-time solution to the drone deployment problem due to its reduced complexity with less than 1 percent degradation in the overall network performance. In addition to the complexity analysis detailed in Sections 3 and 4, we compare the computational complexity of both techniques by measuring the convergence time of each algorithm at the worst case that could happen when several network MTCDs are requesting communication access while they are out-of-coverage of a single placement of the drone BS. The results of this experiment are shown in Fig. 9. In this figure, the time spent to reach a solution is depicted for different problem dimension values n. Needless to mention, the computed time depends on the machine's computational ability and configurations. However, the run time could give an

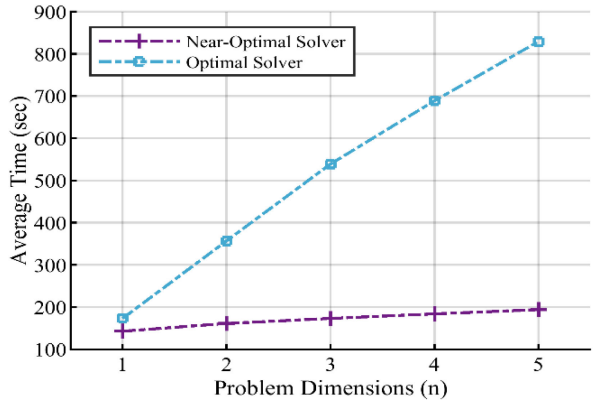

Fig. 9. Average optimization time of the proposed algorithms.

adequate measure or an insight into the algorithmic time complexity and the total number of the computational operations that are required for the proposed optimization solvers regardless of the running host computational power and speed. According to the graph in Fig. 9, we can see that the heuristic algorithm outperforms the exact optimization algorithm with regard to the computational complexity with negligible degradation in the solution quality. The near-optimal solution of the PSO algorithm is shown to have an overall computational complexity that performs in logarithmic time $O(\log n)$, whereas the optimal solution is obtained by much more complex computations of the exact optimization method showing linear time complexity $O(n)$.

6 Conclusion

In this study, we investigated a dynamic drone-mounted BS communication system to serve a dense deployment of different M2M traffic profiles. We introduced two techniques to solve the drone deployment problem; an exact optimization technique and a lower-complexity heuristic technique. The optimization problem was formulated to find the optimal 3D placement of a single drone BS along with the optimal LTE-based resource scheduling to respond to the communication needs of the active MTCDs in the served area. Then, we conducted several simulation experiments to evaluate the functionality of the proposed techniques. To measure the performance of these techniques, we compared them to an optimal deadline-based scheduler that is enabled on a stationary deployment of the drone BS. Using three different system evaluation metrics, namely, the system deadline missing ratio, the system aggregate throughput and the network mean SNR, the proposed exact optimization technique along with the heuristic technique were shown to outperform the static deployment approaches. Furthermore, due to its low complexity, the heuristic-based optimization method succeeds in providing a real-time solution with near-optimal values for the dynamic drone placement and the resource allocation problem.

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